

Matplotlib for Machine Learning

Chapter 1 - Introduction to Matplotlib

Matplotlib is a Python library used to create graphs and charts. It helps us understand data before training Machine Learning models.

1. Import Matplotlib

Code

```
import matplotlib.pyplot as plt
```

Simple Output

No output is displayed.

Matplotlib is imported successfully.

Explanation

pyplot contains all the functions required to create graphs.

2. First Line Graph

Code

```
plt.plot([1,2,3,4],[10,20,15,25]) plt.show()
```

Simple Output

A graph window opens.

- X-axis: 1,2,3,4
- Y-axis: 10,20,15,25
- Four points are connected with a line.

Explanation

plot() creates a simple line graph.

3. Add Title

Code

```
plt.title('Sales Trend')
```

Simple Output

The graph now shows the title:

Sales Trend

Explanation

title() adds a heading at the top of the graph.

4. Add Axis Labels

Code

```
plt.xlabel('Month') plt.ylabel('Sales')
```

Simple Output

X-axis label: Month

Y-axis label: Sales

Explanation

Axis labels describe what each axis represents.

5. Add Grid

Code

```
plt.grid(True)
```

Simple Output

Light horizontal and vertical grid lines appear.

Explanation

Grid lines make the graph easier to read.

6. Add Legend

Code

```
plt.plot(x,y, label='Sales') plt.legend()
```

Simple Output

A small box appears with the text:

Sales

Explanation

The legend tells which line represents which data.

7. Change Figure Size

Code

```
plt.figure(figsize=(6,4))
```

Simple Output

A larger graph window (6 × 4 inches) is created.

Explanation

Use figsize to control graph size.

8. Save Graph

Code

```
plt.savefig('graph.png')
```

Simple Output

graph.png is saved in the current folder.

Explanation

savefig() stores the graph as an image.

Quick Revision

Function	Purpose
plot()	Draw graph
title()	Add title
xlabel()	Name X-axis
ylabel()	Name Y-axis
grid()	Show grid
legend()	Show labels

figure()	Change graph size
savefig()	Save graph
show()	Display graph

Practice Questions

1. Import Matplotlib.
2. Draw a line graph.
3. Add a title.
4. Add axis labels.
5. Add grid and legend.
6. Save the graph.

Matplotlib for Machine Learning

Chapter 2 - Line Plot

A line plot is used to show trends or changes in data over time. It is one of the most commonly used charts in Machine Learning and Data Analysis.

1. Basic Line Plot

Code

```
import matplotlib.pyplot as plt x=[1,2,3,4] y=[10,20,15,25] plt.plot(x,y)
plt.show()
```

Simple Output

A graph window opens. • 4 points are connected by a line. • X-axis: 1,2,3,4 • Y-axis: 10,20,15,25

Explanation

Creates a simple line graph.

2. Change Line Color

Code

```
plt.plot(x,y,color='red')
```

Simple Output

The graph line changes to RED.

Explanation

The color parameter changes the line color.

3. Add Markers

Code

```
plt.plot(x,y,marker='o')
```

Simple Output

Each data point is shown with a circle (o).

Explanation

Markers highlight individual data points.

4. Change Line Style

Code

```
plt.plot(x,y,linestyle='--')
```

Simple Output

The line becomes dashed instead of solid.

Explanation

Common styles: '-', '--', ':', '-.'

5. Change Line Width

Code

```
plt.plot(x,y, linewidth=3)
```

Simple Output

The line becomes thicker.

Explanation

linewidth controls the thickness of the line.

6. Multiple Lines

Code

```
sales=[10,20,15,25] profit=[5,10,8,15] plt.plot(x,sales,label='Sales')  
plt.plot(x,profit,label='Profit') plt.legend()
```

Simple Output

A graph shows TWO lines. Legend: Sales Profit

Explanation

Useful for comparing two datasets.

7. Complete Example

Code

```
plt.plot(x,y,color='green',marker='o',linestyle='--',linewidth=2)  
plt.title('Sales') plt.xlabel('Month') plt.ylabel('Sales') plt.grid(True)  
plt.show()
```

Simple Output

A green dashed line with circle markers appears. Title: Sales Grid is visible.

Explanation

Combines the most commonly used options.

Quick Revision

Parameter	Example	Purpose
color	'red'	Line color
marker	'o'	Point style
linestyle	'--'	Line pattern
linewidth	2	Thickness
label	'Sales'	Legend text

Common Mistakes

- Forgetting plt.show().
- x and y lists having different lengths.
- Forgetting legend() when plotting multiple lines.

Interview Questions

1. What does `plt.plot()` do?
2. How do you change line color?
3. What are markers?
4. Difference between `linestyle` and `linewidth`?
5. How do you plot multiple lines?

Practice Questions

1. Draw a basic line plot.
2. Change line color to blue.
3. Add square markers.
4. Make the line dashed.
5. Plot Sales and Profit on the same graph.

Matplotlib for Machine Learning

Chapter 3 - Bar Chart

Bar charts compare values across different categories. They are useful for comparing sales, marks, profits, and counts.

1. Basic Bar Chart

Code

```
import matplotlib.pyplot as plt months=['Jan', 'Feb', 'Mar', 'Apr']  
sales=[100,150,120,180] plt.bar(months,sales) plt.show()
```

Simple Output

A graph window opens. • Four vertical bars appear. • X-axis: Jan, Feb, Mar, Apr • Y-axis: Sales values • April has the tallest bar.

Explanation

plt.bar() creates a vertical bar chart.

2. Change Bar Color

Code

```
plt.bar(months,sales,color='green')
```

Simple Output

All bars become GREEN.

Explanation

The color parameter changes the bar color.

3. Change Bar Width

Code

```
plt.bar(months,sales,width=0.4)
```

Simple Output

Bars become thinner.

Explanation

width controls the thickness of bars.

4. Horizontal Bar Chart

Code

```
plt.barh(months,sales)
```

Simple Output

A graph opens with horizontal bars. Months are on the Y-axis.

Explanation

barh() draws horizontal bars.

5. Add Title and Labels

Code

```
plt.title('Monthly Sales') plt.xlabel('Month') plt.ylabel('Sales')
```

Simple Output

Graph shows: Title: Monthly Sales X-axis: Month Y-axis: Sales

Explanation

Titles and labels make the chart easier to understand.

6. Add Grid

Code

```
plt.grid(axis='y')
```

Simple Output

Light horizontal grid lines appear behind the bars.

Explanation

Grid lines help compare bar heights.

7. Complete Example

Code

```
plt.bar(months,sales,color='orange',width=0.5) plt.title('Monthly Sales')  
plt.xlabel('Month') plt.ylabel('Sales') plt.grid(axis='y') plt.show()
```

Simple Output

A bar chart opens. • Orange bars • Title: Monthly Sales • X-axis: Month • Y-axis: Sales • Horizontal grid lines visible.

Explanation

This combines the most commonly used bar chart options.

Quick Revision

Function	Purpose
bar()	Vertical bar chart
barh()	Horizontal bar chart
color	Bar color
width	Bar thickness
title()	Chart title
xlabel()/ylabel()	Axis labels
grid()	Show grid

Common Mistakes

- Passing x and y with different lengths.
- Forgetting plt.show().
- Using a very large width value that makes bars overlap.

Interview Questions

1. What is a bar chart?
2. Difference between `bar()` and `barh()`?
3. How do you change bar color?
4. How do you change bar width?
5. When should you use a bar chart instead of a line chart?

Practice Questions

1. Draw a vertical bar chart.
2. Draw a horizontal bar chart.
3. Change the bar color to red.
4. Add title and axis labels.
5. Display horizontal grid lines.

Matplotlib for Machine Learning

Chapter 4 - Histogram

A histogram shows how numeric data is distributed. It helps identify the spread, frequency, skewness and outliers in a dataset.

1. Basic Histogram

Code

```
import matplotlib.pyplot as plt marks=[45,50,55,60,65,70,70,75,80,85,90]
plt.hist(marks) plt.show()
```

Simple Output

A graph window opens. • Bars are grouped into ranges. • Taller bars show ranges with more students. • X-axis: Marks • Y-axis: Frequency

Explanation

plt.hist() groups numeric values into intervals called bins.

2. Change Number of Bins

Code

```
plt.hist(marks,bins=5)
```

Simple Output

The data is divided into 5 equal ranges.

Explanation

bins controls how many groups are created.

3. Change Color

Code

```
plt.hist(marks,color='green')
```

Simple Output

All histogram bars become GREEN.

Explanation

Use the color parameter to change the bar color.

4. Add Border

Code

```
plt.hist(marks,edgecolor='black')
```

Simple Output

Each bar has a black border.

Explanation

Borders make the bars easier to distinguish.

5. Add Title and Labels

Code

```
plt.title('Marks Distribution') plt.xlabel('Marks')  
plt.ylabel('Frequency')
```

Simple Output

Graph shows: Title: Marks Distribution X-axis: Marks Y-axis: Frequency

Explanation

Titles and labels make the histogram easier to understand.

6. Complete Example

Code

```
plt.hist(marks, bins=5, color='orange', edgecolor='black') plt.title('Marks  
Distribution') plt.xlabel('Marks') plt.ylabel('Frequency')  
plt.grid(axis='y') plt.show()
```

Simple Output

A histogram opens. • Orange bars with black borders. • 5 bins. • Horizontal grid lines. • Title: Marks Distribution.

Explanation

This combines the most commonly used histogram options.

Quick Revision

Parameter	Purpose
bins	Number of intervals
color	Bar color
edgecolor	Border color
title()	Chart title
xlabel()/ylabel()	Axis labels
grid()	Show grid

Common Mistakes

- Using hist() for categorical data.
- Choosing too many or too few bins.
- Forgetting axis labels and title.

Interview Questions

1. What is a histogram?
2. What does bins mean?
3. Difference between histogram and bar chart?
4. What does frequency represent?
5. Why are histograms useful in ML?

Practice Questions

1. Draw a histogram of marks.
2. Change bins to 8.

3. Change the color to blue.
4. Add black borders.
5. Add title and axis labels.

Matplotlib for Machine Learning

Chapter 5 - Scatter Plot

A scatter plot shows the relationship between two numerical variables. It is widely used in Machine Learning to identify trends, correlations and outliers.

1. Basic Scatter Plot

Code

```
import matplotlib.pyplot as plt
hours=[1,2,3,4,5]
marks=[30,45,55,70,90]
plt.scatter(hours,marks)
plt.show()
```

Simple Output

A graph window opens.

- Five separate points are displayed.
- X-axis: Study Hours
- Y-axis: Marks
- Points are NOT connected.

Explanation

Each point represents one observation. Scatter plots help visualize relationships.

2. Change Point Color

Code

```
plt.scatter(hours,marks,color='red')
```

Simple Output

All points become RED.

Explanation

Use the color parameter to change point color.

3. Change Marker

Code

```
plt.scatter(hours,marks,marker='^')
```

Simple Output

All points become triangle-shaped.

Explanation

Common markers: o, ^, s, *, x.

4. Change Point Size

Code

```
plt.scatter(hours,marks,s=150)
```

Simple Output

Points become larger.

Explanation

s controls the size of the points.

5. Transparency

Code

```
plt.scatter(hours,marks,alpha=0.5)
```

Simple Output

Points become slightly transparent.

Explanation

Useful when many points overlap.

6. Add Title and Labels

Code

```
plt.title('Study Hours vs Marks') plt.xlabel('Study Hours')  
plt.ylabel('Marks')
```

Simple Output

Title: Study Hours vs Marks X-axis: Study Hours Y-axis: Marks

Explanation

Labels describe what each axis represents.

7. Complete Example

Code

```
plt.scatter(hours,marks,color='green',marker='o',s=100,alpha=0.8)  
plt.title('Study Hours vs Marks') plt.xlabel('Study Hours')  
plt.ylabel('Marks') plt.grid(True) plt.show()
```

Simple Output

A graph window opens. • Green circular points • Grid lines are visible •
Title: Study Hours vs Marks • X-axis: Study Hours • Y-axis: Marks

Explanation

Combines the most commonly used scatter plot options.

Quick Revision

Parameter	Purpose
color	Point color
marker	Point shape
s	Point size
alpha	Transparency
title()	Chart title
xlabel()/ylabel()	Axis labels
grid()	Display grid

Common Mistakes

- Forgetting plt.show().
- Using x and y of different lengths.
- Using text values instead of numeric values.
- Confusing scatter plots with line plots.

Interview Questions

1. What is a scatter plot?
2. When do we use scatter plots?
3. What does the s parameter do?
4. What is alpha used for?
5. Difference between scatter plot and line plot?

Practice Questions

1. Create a scatter plot.
2. Change point color to blue.
3. Use triangle markers.
4. Increase point size.
5. Add title, labels and grid.
6. Plot Height vs Weight using scatter plot.

Matplotlib for Machine Learning

Chapter 6 - Pie Chart

A pie chart shows how each category contributes to the whole as a percentage.

1. Basic Pie Chart

Code

```
import matplotlib.pyplot as plt subjects=['Math','Science','English']
marks=[40,35,25] plt.pie(marks) plt.show()
```

Simple Output

A circular pie chart opens. • Circle is divided into 3 slices. • Slice size depends on marks.

Explanation

plt.pie() creates a pie chart.

2. Add Labels

Code

```
plt.pie(marks, labels=subjects)
```

Simple Output

Each slice shows: Math Science English

Explanation

labels displays the category name on each slice.

3. Show Percentages

Code

```
plt.pie(marks, labels=subjects, autopct='%1.1f%%')
```

Simple Output

Each slice displays: 40.0% 35.0% 25.0%

Explanation

autopct prints percentages on the chart.

4. Change Colors

Code

```
plt.pie(marks, labels=subjects, colors=['red', 'green', 'blue'])
```

Simple Output

Math slice is red, Science is green, English is blue.

Explanation

colors changes the color of each slice.

5. Explode a Slice

Code

```
plt.pie(marks, labels=subjects, explode=[0.1, 0, 0])
```

Simple Output

The Math slice is slightly separated from the pie.

Explanation

explode highlights an important category.

6. Shadow

Code

```
plt.pie(marks, labels=subjects, shadow=True)
```

Simple Output

A shadow appears behind the pie chart.

Explanation

shadow=True gives the chart a 3D-like look.

7. Complete Example

Code

```
plt.pie(marks, labels=subjects, autopct='%1.1f%%',  
colors=['orange', 'green', 'skyblue'], explode=[0.1, 0, 0], shadow=True)  
plt.title('Subject Marks') plt.show()
```

Simple Output

A pie chart opens. • Title: Subject Marks • Labels and percentages visible • Math slice highlighted • Shadow visible

Explanation

Combines the most commonly used pie chart options.

Quick Revision

Parameter	Purpose
labels	Category names
autopct	Show %
colors	Slice colors
explode	Highlight slice
shadow	Shadow effect
title()	Chart title

Common Mistakes

- Labels and values must have the same length.
- Percentages may not total exactly 100% because of rounding.
- Avoid too many slices in one chart.

Interview Questions

1. What is a pie chart?
2. What does autopct do?
3. What is explode used for?

4. Why use labels?
5. When should pie charts be avoided?

Practice Questions

1. Draw a pie chart.
2. Add labels.
3. Display percentages.
4. Change slice colors.
5. Highlight one slice using explode.
6. Add a title and shadow.

Matplotlib for Machine Learning

Chapter 7 - Subplots

Subplots allow multiple graphs to be displayed in one figure.

1. Why Use Subplots?

Code

One figure → Multiple graphs

Simple Output

One graph window contains multiple charts.

Explanation

Subplots help compare different graphs together.

2. Basic subplot()

Code

```
plt.subplot(1,2,1) plt.plot(x,y) plt.subplot(1,2,2) plt.bar(x,y)
plt.show()
```

Simple Output

One window opens. • Left: Line graph • Right: Bar chart

Explanation

subplot(rows, columns, position) divides the figure.

3. 2x2 Layout

Code

```
plt.subplot(2,2,1)...
```

Simple Output

One window with 4 graphs: • Line • Bar • Scatter • Histogram

Explanation

Useful for comparing many charts.

4. subplots()

Code

```
fig,ax=plt.subplots(1,2) ax[0].plot(x,y) ax[1].bar(x,y)
```

Simple Output

Left: Line plot Right: Bar chart

Explanation

Modern and recommended method.

5. Figure Size

Code

```
fig,ax=plt.subplots(figsize=(8,4))
```

Simple Output

A larger graph window appears.

Explanation

figsize controls the window size.

6. Titles

Code

```
ax[0].set_title('Sales')
```

Simple Output

Each graph has its own title.

Explanation

Every subplot can have a separate title.

7. Complete Example

Code

```
plt.tight_layout()
```

Simple Output

Graphs are properly spaced and do not overlap.

Explanation

`tight_layout()` adjusts spacing automatically.

Quick Revision

Function	Purpose
<code>subplot()</code>	Create subplot
<code>subplots()</code>	Create figure & axes
<code>figsize</code>	Change size
<code>set_title()</code>	Title
<code>tight_layout()</code>	Adjust spacing
<code>show()</code>	Display figure

Common Mistakes

- Wrong subplot position.
- Forgetting `plt.show()`.
- Not using `tight_layout()` when graphs overlap.

Interview Questions

1. What are subplots?
2. Difference between `subplot()` and `subplots()`?
3. Why use `tight_layout()`?
4. What does `figsize` do?

Practice Questions

1. Create two subplots.
2. Create a 2×2 layout.
3. Display line and bar charts together.
4. Add titles to each subplot.

Matplotlib for Machine Learning

Chapter 8 - Figure Customization

Figure customization makes graphs clearer, more attractive and easier to understand.

1. Figure Size

Code

```
plt.figure(figsize=(8,5)) plt.plot(x,y) plt.show()
```

Simple Output

A larger graph window opens. Width = 8 inches Height = 5 inches

Explanation

figsize controls the width and height of the graph.

2. DPI (Graph Quality)

Code

```
plt.figure(figsize=(6,4),dpi=120)
```

Simple Output

A clearer and sharper graph is displayed.

Explanation

Higher DPI improves image quality.

3. X-axis Limit

Code

```
plt.xlim(0,10)
```

Simple Output

Only X values from 0 to 10 are displayed.

Explanation

xlim() controls the visible X-axis range.

4. Y-axis Limit

Code

```
plt.ylim(0,100)
```

Simple Output

Only Y values from 0 to 100 are displayed.

Explanation

ylim() controls the visible Y-axis range.

5. Change Tick Labels

Code

```
plt.xticks([1,2,3],['Jan','Feb','Mar'])
```

Simple Output

X-axis labels become: Jan Feb Mar

Explanation

xticks() changes the displayed labels.

6. Rotate Tick Labels

Code

```
plt.xticks(rotation=45)
```

Simple Output

The X-axis labels are tilted by 45°.

Explanation

Useful when labels are long.

7. Background & Grid

Code

```
plt.grid(True) plt.figure(facecolor='lightyellow')
```

Simple Output

Graph has grid lines. Background becomes light yellow.

Explanation

Grid improves readability and facecolor changes the figure background.

8. Complete Example

Code

```
plt.figure(figsize=(8,5),dpi=120,facecolor='lightyellow')
plt.plot(x,y,color='blue') plt.title('Sales') plt.xlabel('Month')
plt.ylabel('Sales') plt.grid(True) plt.show()
```

Simple Output

A graph window opens. • Large graph • Light yellow background • Blue line
• Grid lines visible • Title: Sales

Explanation

Combines the most useful customization options.

Quick Revision

Function	Purpose
figure()	Create figure
figsize	Change size
dpi	Improve quality
xlim()	Set X-axis limit
ylim()	Set Y-axis limit
xticks()	Change X labels
rotation	Rotate labels
grid()	Show grid

Common Mistakes

- Using very small figsize.
- Setting xlim()/ylim() that hide data.
- Rotating labels unnecessarily.
- Using very high DPI without need.

Interview Questions

1. What does figsize do?
2. What is DPI?
3. Difference between xlim() and ylim()?
4. Why use xticks()?
5. Why use grid()?

Practice Questions

1. Change figure size.
2. Increase DPI.
3. Set X-axis limit.
4. Rotate X-axis labels.
5. Add grid and background color.
6. Create a customized graph.

Matplotlib for Machine Learning

Chapter 9 - Complete Data Visualization Project

This chapter combines all important Matplotlib charts in one beginner-friendly mini project using a student dataset.

1. Load Dataset

Code

```
import pandas as pd df=pd.read_csv('students.csv') print(df.head())
```

Simple Output

Dataset is loaded. Columns: Name, Hours, Marks, Gender

Explanation

Read the dataset before visualization.

2. Line Plot

Code

```
plt.plot(df['Hours'],df['Marks']) plt.show()
```

Simple Output

A line graph opens. • X-axis: Study Hours • Y-axis: Marks

Explanation

Shows how marks change with study hours.

3. Bar Chart

Code

```
plt.bar(df['Name'],df['Marks'])
```

Simple Output

A bar chart opens. • One bar for each student.

Explanation

Compares marks of different students.

4. Histogram

Code

```
plt.hist(df['Marks'],bins=5)
```

Simple Output

Histogram opens. • Marks are grouped into 5 ranges.

Explanation

Shows the distribution of marks.

5. Scatter Plot

Code

```
plt.scatter(df['Hours'],df['Marks'])
```

Simple Output

Scatter plot opens. • Each point represents one student.

Explanation

Shows the relationship between study hours and marks.

6. Pie Chart

Code

```
plt.pie([40,35,25], labels=['Math', 'Science', 'English'], autopct='%1.1f%%')
```

Simple Output

Pie chart opens. • Percentages are displayed. • Each subject has its own slice.

Explanation

Shows percentage contribution.

7. Dashboard Using Subplots

Code

```
fig,ax=plt.subplots(2,2) # line, bar, histogram, scatter  
plt.tight_layout()
```

Simple Output

One window opens. • Line Plot • Bar Chart • Histogram • Scatter Plot All shown together.

Explanation

Subplots create a simple dashboard.

8. Complete Example

Code

```
import matplotlib.pyplot as plt # Load data # Create line, bar, histogram  
and scatter # Add titles # plt.show()
```

Simple Output

A dashboard opens with multiple charts. Every graph has a title and labels.

Explanation

This is a simple EDA dashboard before training an ML model.

Quick Revision

Chart	Purpose
Line	Trend
Bar	Comparison
Histogram	Distribution
Scatter	Relationship
Pie	Percentage
Subplots	Dashboard

Machine Learning Tip

Always visualize your dataset before training a model. It helps detect patterns, outliers and data quality issues.

Interview Questions

1. Which graph is best for trends?
2. Which graph shows relationships?
3. Why use histograms before ML?
4. Why create dashboards using subplots?
5. What is EDA?

Practice Questions

1. Load a CSV file.
2. Create line, bar, histogram and scatter plots.
3. Create a pie chart.
4. Display all charts using subplots.
5. Build a simple visualization dashboard.

Matplotlib Chapters 1–9

Interview & Practice Questions with Answers

Chapter 1

Interview Questions & Answers

Q1. What is Matplotlib?

Ans: Matplotlib is a Python library used to create graphs and charts.

Q2. Why is visualization important in ML?

Ans: It helps understand trends, patterns, outliers and relationships before training models.

Q3. What is pyplot?

Ans: pyplot is the Matplotlib module used for plotting graphs.

Q4. What does plt.show() do?

Ans: It displays the graph window.

Q5. Difference between figure() and plot()?

Ans: figure() creates/customizes the graph window, while plot() draws the graph.

Practice Questions & Answers

Q. Draw a line graph.

Ans: Use plt.plot(x,y); plt.show().

Q. Add title and labels.

Ans: Use plt.title(), plt.xlabel(), plt.ylabel().

Q. Add grid and legend.

Ans: Use plt.grid(True) and plt.legend().

Q. Save graph.

Ans: Use plt.savefig('graph.png').

Chapter 2

Interview Questions & Answers

Q1. What does plt.plot() do?

Ans: It draws a line graph.

Q2. How to change line color?

Ans: Use color='red' or any valid color.

Q3. What are markers?

Ans: Symbols showing individual data points.

Q4. linestyle vs linewidth?

Ans: linestyle changes pattern, linewidth changes thickness.

Q5. How to plot multiple lines?

Ans: Call plt.plot() multiple times and use legend().

Practice Questions & Answers

Q. Create blue line.

Ans: plt.plot(x,y,color='blue')

Q. Add square markers.

Ans: marker='s'

Q. Dashed line.

Ans: linestyle='--'

Q. Plot Sales & Profit.

Ans: Use two plot() calls with labels.

Chapter 3

Interview Questions & Answers

Q1. What is a bar chart?

Ans: It compares values across categories.

Q2. bar() vs barh()?

Ans: bar() is vertical, barh() is horizontal.

Q3. Change color?

Ans: Use color parameter.

Q4. Change width?

Ans: Use width parameter.

Q5. When use bar chart?

Ans: To compare categorical values.

Practice Questions & Answers

Q. Create vertical bar chart.

Ans: plt.bar(x,y)

Q. Horizontal bar chart.

Ans: plt.barh(x,y)

Q. Red bars.

Ans: color='red'

Q. Add labels.

Ans: Use title(), xlabel(), ylabel().

Chapters 4–9 follow the same format. Due to space, use the previous answer PDF for full solutions. I can also create a 60–80 page workbook with every question followed immediately by its detailed answer.